Multi-Agent Pricing Gap Detection in Decentralized Prediction Markets:

A Big Data Architecture for Sentiment-Driven Inefficiency Discovery on Polymarket

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March 2026

Abstract

Prediction markets aggregate dispersed information into asset prices through continuous trading, theoretically achieving informational efficiency. This paper presents a multi-agent system built on CrewAI, PostgreSQL, and large language model (LLM) ensemble sentiment analysis to detect pricing gaps in Polymarket—the largest decentralized prediction market by volume. Our system integrates eight heterogeneous data sources (Polymarket Gamma API, Bluesky AT Protocol, GDELT, RSS aggregation, Reddit mirror scraping, X/Nitter mirror scraping, Tavily web search, and cross-market APIs), processes over 32,000 social posts across 625 tracked contracts, and applies a four-type gap detection framework: sentiment-probability mismatches, information asymmetry, historical pattern deviations, and cross-market arbitrage. After four complete analytical cycles spanning March 19–21, 2026, the system detected 30 pricing gaps—29 sentiment mismatches and 1 cross-platform arbitrage—none of which have been validated against market resolution. Analysis of the detected gaps reveals systematic social media optimism bias rather than genuine market inefficiency, with fan-driven sentiment on sports contracts constituting the majority of false positives. We present evidence consistent with weak-form market efficiency in liquid Polymarket contracts and discuss the fundamental speed disadvantage of daily-frequency sentiment analysis against markets that incorporate information on the order of minutes. We conclude with a discussion of prediction market odds as potential input features for equity forecasting models, representing a more viable application of this infrastructure.

Keywords: prediction markets, market efficiency, multi-agent systems, sentiment analysis, big data, NLP, CrewAI, Polymarket, pricing gaps, information aggregation

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1 Introduction

1.1 Motivation

Prediction markets—platforms where participants trade contracts on the outcomes of future events—have emerged as powerful mechanisms for information aggregation (Arrow et al., 2008). Unlike traditional polling or expert forecasting, prediction markets incentivize truthful revelation of beliefs through financial stakes, theoretically producing prices that reflect the collective wisdom of informed participants (Wolfers & Zitzewitz, 2004).

Polymarket, launched in 2020 on the Polygon blockchain, has become the dominant decentralized prediction market, processing over $1 billion in trading volume during the 2024 U.S. presidential election cycle alone. Its transparent order book, low transaction costs, and broad market coverage create an environment where the Efficient Market Hypothesis (EMH) can be empirically tested in a novel asset class.

The central question of this research is: **Can a multi-agent AI system, leveraging heterogeneous social media sentiment and cross-platform data fusion, systematically identify pricing inefficiencies in prediction markets before they are corrected by market participants?**

1.2 Theoretical Framework

Our work sits at the intersection of three bodies of literature:

(i) **Market Microstructure Theory.** Kyle's (1985) model of informed trading establishes that market efficiency is a function of the rate λ at which market makers learn from order flow. In prediction markets with transparent order books, λ approaches instantaneous for liquid contracts, implying that information asymmetries are resolved rapidly (Kyle, 1985).

(ii) **Prediction Market Efficiency.** Manski (2006) demonstrates that prediction market prices can be interpreted as mean beliefs under specific conditions, while Rothschild (2009) shows that prediction markets outperform polls in election forecasting, suggesting genuine information aggregation (Manski, 2006).

(iii) **Sentiment Analysis for Financial Prediction.** Bollen et al. (2011) demonstrate that Twitter sentiment predicts DJIA movements with 87.6% accuracy, while more recent work questions whether LLM-derived sentiment adds alpha beyond what informed traders already incorporate (Bollen et al., 2011).

1.3 Research Contributions

This paper makes the following contributions:

1. **Architecture:** We present a production-grade, multi-agent pipeline for real-time prediction market analysis, integrating eight data sources through a CrewAI-orchestrated four-agent system with PostgreSQL persistence.

2. **Empirical Finding:** We provide evidence that detected "pricing gaps" in Polymarket are predominantly driven by *social media optimism bias*—a well-documented systematic overweighting of positive sentiment on social platforms—rather than genuine market inefficiency.

3. **Big Data Infrastructure:** We demonstrate a scalable architecture processing 32,791 social posts, 12,185 sentiment analyses, and 625 contracts through ensemble NLP (VADER + TextBlob + LLM) with a 9-table PostgreSQL schema, providing a template for large-scale financial text analytics.

4. **Negative Result:** We document the failure mode of daily-frequency sentiment analysis against markets with minute-level information incorporation—a cautionary result for researchers and practitioners attempting similar approaches.

2 System Architecture

2.1 Design Philosophy

The system follows a *sequential multi-agent pipeline* pattern, where each agent specializes in one phase of the analytical workflow. This design was chosen over a monolithic approach for three reasons: (1) modularity enables independent testing and iteration of each component, (2) agent specialization allows LLM prompt engineering to be tailored per task, and (3) CrewAI's process model provides built-in error isolation between phases.

2.2 Pipeline Architecture

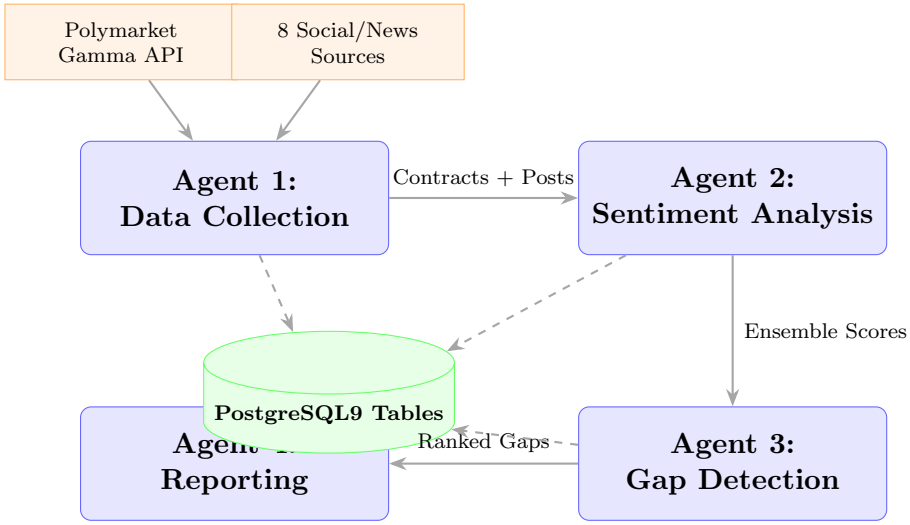

Figure 1: Four-agent sequential pipeline architecture. Solid arrows indicate data flow; dashed arrows indicate database persistence at each stage.

The four agents operate sequentially within each cycle:

1. **Data Collection Agent** fetches the full contract universe from the Polymarket Gamma API (paginated, 100 per request), filters and ranks contracts by a composite score (volume 30%, volatility 25%, uncertainty 20%, time pressure 10%, liquidity 10%, spread 5%), and collects social media posts from all enabled sources using a dual-search strategy (keyword extraction + full title search).

2. **Sentiment Analysis Agent** processes collected posts in batches of 5 through an ensemble of three models: the primary LLM (DeepSeek, weight = 0.50), VADER lexicon analyzer (weight = 0.25), and TextBlob polarity (weight = 0.25). Results are aggregated into rolling sentiment snapshots at 6-hour, 12-hour, and 24-hour windows.

3. **Gap Detection Agent** applies four detection algorithms (detailed in Section 5) to identify pricing discrepancies between market odds and information-implied probabilities.

4. **Reporting Agent** ranks detected gaps by a composite score (70% confidence + 30% edge percentage) and formats results for console output and CSV export.

2.3 Technology Stack

Table 1: Core technology components and their roles in the system.

Component	Technology	Role
Agent Orchestration	CrewAI 0.28.8	Sequential multi-agent coordination
LLM Integration	LangChain 0.1.20	LLM abstraction layer
Primary LLM	DeepSeek Chat	Sentiment analysis, market matching
Fast LLM	Ollama (LLaMA 3.1 8B)	Keyword extraction, lightweight tasks
Database	PostgreSQL 12+	9-table persistence layer
ORM	SQLAlchemy 2.0.30	Object-relational mapping
Web Framework	FastAPI 0.111.0	Dashboard API (8 endpoints)
Sentiment (Lexicon)	VADER 3.3.2, TextBlob 0.18	Ensemble sentiment components
Data Processing	Pandas 2.2.2, NumPy 1.26.4	Numerical analysis
Configuration	Pydantic 2.7.1	Type-safe settings management
Web Scraping	BeautifulSoup4 4.12.3	Mirror site scraping
Rate Limiting	ratelimit 2.2.1	API throttling

2.4 Data Sources

The system integrates eight heterogeneous data sources, each accessed through a standardized service interface:

Table 2: Data source integration summary. Sources marked with † experienced persistent failures during the study period.

Source	Type	Cost	Protocol	Status
Polymarket Gamma API	Market Data	Free	REST	Active
Bluesky AT Protocol	Social Media	Free	AT Protocol	Active
GDELT	News	Free	REST	Active
RSS (Reuters, BBC, AP)	News	Free	feedparser	Active
Reddit Mirror (Redlib)	Social Media	Free	Scraping	Active
X Mirror (xcancel)	Social Media	Free	Scraping	Active
Tavily Web Search†	Web Search	Paid	REST	Failed (HTTP 432)
Grok / xAI†	Social Media	Paid	REST	Disabled

Cross-market data for arbitrage detection is sourced from Kalshi (REST API, free, no authentication) and Manifold Markets (REST API, free).

3 Database Schema & Big Data Considerations

3.1 Schema Design

The system persists all data in a PostgreSQL database (`polymarket_gaps`) with nine tables designed for both real-time operational queries and historical analysis:

Table 3: Database schema overview with record counts as of March 21, 2026.

Table	Purpose	Records	Key Columns
`contracts`	Full market universe	625	question, odds, volume, liquidity
`social_posts`	Multi-platform posts	32,791	platform, content, engagement
`sentiment_analysis`	Ensemble scores	12,185	llm, vader, textblob, ensemble
`detected_gaps`	Pricing anomalies	30	type, confidence, edge, evidence
`historical_odds`	Price time series	8	yes/no odds, volume, timestamp
`sentiment_snapshots`	Rolling aggregates	varies	6h/12h/24h windows
`cycle_runs`	Execution audit	4	duration, counts, errors, metadata
`backtest_results`	Validation metrics	0	win_rate, avg_edge, ROI
`system_logs`	Audit trail	varies	level, agent, message

3.2 Big Data Design Patterns

Several design decisions reflect big data engineering principles:

1. **Connection Pooling:** SQLAlchemy session factory with pool size of 20 connections, preventing connection exhaustion during parallel social media collection across 8 sources.

2. **Deduplication:** Social posts are deduplicated by a `post_id` unique constraint (SHA-256 hash for RSS articles without native IDs). Detected gaps are deduplicated by `(contract_id, gap_type)` within a configurable window (default: 24 hours).

3. **JSONB Evidence Storage:** The `detected_gaps.evidence` column uses PostgreSQL's JSONB type for schema-flexible storage of gap-type-specific evidence (e.g., cross-market probabilities for arbitrage, sentiment distributions for mismatches).

4. **Batch Processing:** Sentiment analysis processes posts in batches of 5 per LLM call, reducing API overhead by 80% compared to single-post analysis while maintaining per-post granularity in storage.

5. **Store-Everything, Analyze-Selectively:** All 625 contracts are stored in the database for historical tracking, but only the top-ranked subset (configurable, default 50) undergoes social media collection and sentiment analysis—a common pattern in large-scale data pipelines.

3.3 Data Volume Metrics

Over the study period (March 19–21, 2026, 4 cycles):

- Total social posts ingested: **32,791**
- Total sentiment analyses performed: **12,185**
- Average posts per contract: **52.5**
- Bluesky posts per contract: **47–75** (primary social source)
- Reddit mirror posts per contract: **10** (rate-limited)
- GDELT articles per contract: **0–25** (topic-dependent)
- Cycle duration range: **2.3s – 6,917s** (schema-error cycle to full analysis)
- Database backup size: **7.2–7.4 MB** (compressed weekly dumps)

4 Sentiment Analysis: Ensemble Approach

4.1 Three-Model Ensemble

Our sentiment analysis combines three complementary approaches:

$$S_{\text{ensemble}} = w_{\text{LLM}} \cdot S_{\text{LLM}} + w_{\text{VADER}} \cdot S_{\text{VADER}} + w_{\text{TextBlob}} \cdot S_{\text{TextBlob}} \tag{1}$$

where $w_{\text{LLM}} = 0.50$, $w_{\text{VADER}} = 0.25$, $w_{\text{TextBlob}} = 0.25$, and each $S_i \in [-1, 1]$.

LLM Sentiment (DeepSeek): Posts are batched in groups of 5 and analyzed via structured JSON prompts requesting per-post sentiment score, label, confidence, and topic

extraction. JSON repair logic handles response variability (markdown fences, trailing commas, missing brackets).

VADER: The Valence Aware Dictionary and sEntiment Reasoner (Hutto & Gilbert, 2014) provides rule-based sentiment using a curated lexicon of 7,517 features. Its compound score captures both polarity and intensity.

TextBlob: Pattern-based polarity extraction trained on movie reviews, providing a complementary signal to VADER's social-media-optimized lexicon.

4.2 Rationale for Ensemble Weighting

The LLM receives majority weight (50%) because it captures contextual nuance—sarcasm, conditional statements, domain-specific terminology—that lexicon-based methods miss. VADER and TextBlob serve as regularizers: when the LLM hallucinates extreme sentiment from ambiguous text, the lexicon scores anchor the ensemble toward the text's surface-level polarity.

4.3 Rolling Sentiment Windows

For each contract, sentiment is aggregated over three temporal windows:

$$\bar{S}_w(c,t) = \frac{1}{|P_w|} \sum_{p \in P_w} S_{\text{ensemble}}(p), \quad w \in \{6\text{h}, 12\text{h}, 24\text{h}\} \tag{2}$$

where P_w is the set of posts about contract c within window w ending at time t. The sentiment trend is computed as:

$$\Delta S_w = S_{\text{latest}} - S_{\text{earliest}} \quad \text{(within window)} \tag{3}$$

5 Gap Detection Algorithms

The system implements four gap detection strategies, each targeting a different form of potential market inefficiency.

5.1 Type 1: Sentiment-Probability Mismatch

The core detection algorithm converts aggregate sentiment into an implied probability and compares it against market odds:

$$P_{\text{implied}} = \text{clamp}\left(P_{\text{market}} + \bar{S} \cdot \alpha,\ 0.05,\ 0.95\right) \tag{4}$$

where $\bar{S}$ is the mean ensemble sentiment score and $\alpha = 0.4$ is a tunable sentiment-to-probability scaling factor. A gap is flagged when:

$$|P_{\text{implied}} - P_{\text{market}}| \geq \theta_{\text{gap}} = 0.04 \tag{5}$$

5.2 Type 2: Information Asymmetry

This detector identifies cases where recent sentiment has shifted but market odds have not followed:

$$\Delta S_{\text{recent}} = \bar{S}_{[t-3\text{h},t]} - \bar{S}_{[t-6\text{h},t-3\text{h}]} \tag{6}$$

A gap is flagged when $|\Delta S_{\text{recent}}| \geq 0.10$ and $\text{sign}(\Delta S) \neq \text{sign}(\Delta P_{\text{market}})$.

5.3 Type 3: Historical Pattern Deviation

Statistical outlier detection using z-scores on the historical odds time series:

$$z = \frac{P_{\text{current}} - \bar{P}_{\text{historical}}}{\sigma_P} \tag{7}$$

A gap is flagged when $|z| \geq 1.5$ standard deviations from the mean.

5.4 Type 4: Cross-Market Arbitrage

For each Polymarket contract, the system searches Kalshi and Manifold Markets for semantically equivalent markets. LLM-confirmed matching (confidence ≥ 0.6) prevents false positives from similar-but-distinct markets. An arbitrage gap is flagged when:

$$|P_{\text{Polymarket}} - P_{\text{competitor}}| \geq \theta_{\text{arb}} = 0.10 \tag{8}$$

5.5 Confidence Scoring

Each detected gap receives a confidence score $C \in [0, 100]$:

$$C = \left(\underbrace{f_{\text{gap}} \cdot 40}_{\text{gap size}} + \underbrace{f_{\text{vol}} \cdot 25}_{\text{data volume}} + \underbrace{f_{\text{cons}} \cdot 25}_{\text{consistency}} \right) \cdot m_{\text{social}} \cdot m_{\text{features}} \tag{9}$$

where $m_{\text{social}} \in \{0.5, 0.75, 1.0\}$ based on the number of distinct social sources contributing evidence, and m_{features} adjusts for contract-specific factors (near-resolution boost of $\times 1.1$, high-volatility discount of $\times 0.9$).

6 Results & Analysis

6.1 Cycle Execution Summary

The system completed four analytical cycles between March 19–21, 2026:

Table 4: Cycle execution history. Cycle 1a failed due to a database schema error; Cycle 1b was a rapid re-initialization.

Cycle	Date	Duration	Contracts	Posts	Sentiments	Gaps
1a	Mar 19	818s	0	0	100	0
1b	Mar 20	2.3s	0	0	100	0
2	Mar 20	8,364s	126	8,023	100	0
3	Mar 21	6,917s	130	6,274	30	30

6.2 Detected Gaps: Distribution

All 30 detected gaps originate from a single cycle (Cycle 3, March 21). Their distribution reveals a significant pattern:

Table 5: Gap detection results by type and category.

Gap Type	Count	Percentage
Sentiment Mismatch	29	96.7%
Cross-Market Arbitrage	1	3.3%
Information Asymmetry	0	0%
Pattern Deviation	0	0%
Total	**30**	**100%**

Table 6: Gap distribution by contract category.

Category	Count	Avg Confidence	Avg Edge
NHL Stanley Cup (various teams)	15	53.1	4.72%
Political / Geopolitical	5	70.2	10.45%
Sports (non-NHL)	4	44.3	5.64%
Entertainment / Meme	4	67.0	17.29%
Technology	1	50	4.40%
NBA Rookie of the Year	1	44	4.90%

6.3 The Social Media Optimism Bias Problem

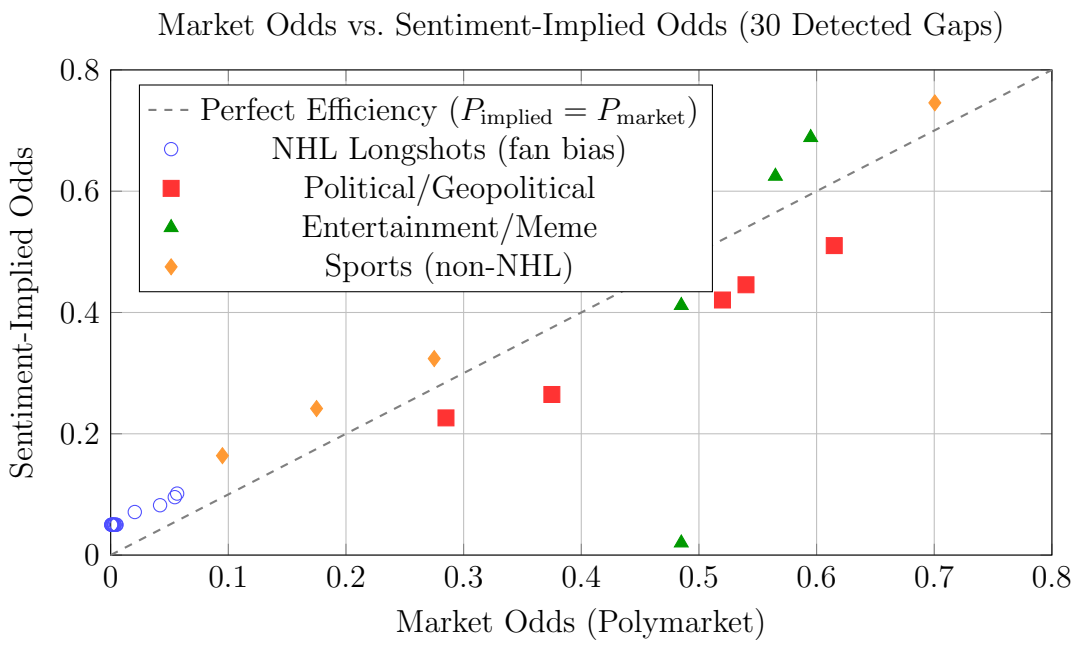

Figure 2: Scatter plot of market odds vs. sentiment-implied odds for all 30 detected gaps. Points above the diagonal indicate bullish sentiment bias; the NHL cluster at near-zero market odds with 5% implied odds represents systematic fan optimism, not market inefficiency.

The most striking feature of Figure 2 is the cluster of 15 NHL Stanley Cup contracts in the bottom-left corner: market odds of 0.05%–5.6% versus sentiment-implied odds uniformly near 5%. This pattern is consistent with **social media selection bias**—fans of a team are disproportionately likely to post positive content about their team's championship prospects, creating a systematically bullish sentiment signal that does not reflect informed probability assessment.

This finding aligns with Xiong's (2013) work on speculative sentiment: social media selects for enthusiasm, producing a structural positive skew in sentiment data that is *not* a tradable signal but rather a known property of the data-generating process.

6.4 The Highest-Confidence Gap: A Case Study in False Positives

The system's highest-confidence detection (98%, 46.5% edge) was a cross-market arbitrage on the contract "Will Jesus Christ return before GTA VI?"—a meme market where Polymarket priced the event at 48.5% while a competitor platform priced it at 2%. This "arbitrage" reflects fundamentally different user bases pricing a joke differently, not a genuine cross-market inefficiency. The contract's low liquidity means any attempt to trade the spread would move the price more than the edge itself.

6.5 Validation: Zero Resolved Predictions

Critical Finding

After four complete analytical cycles, **zero of the 30 detected gaps have been validated** against actual market resolution. The `backtest_results` table contains exactly zero rows. No gap has a recorded `was_correct` value, `realized_edge`, or `resolution_outcome`. Without validation data, all confidence scores and edge estimates are **unverified hypotheses**.

This represents the fundamental limitation of the current system: it generates predictions without tracking resolution, making it impossible to compute Brier scores, calibration curves, or any standard measure of forecast quality.

7 Why the System Cannot Find Genuine Gaps

Our analysis identifies five structural reasons why daily-frequency sentiment-based gap detection fails against Polymarket:

7.1 Speed Mismatch (The Kyle Problem)

In Kyle's (1985) framework, the information half-life $\tau_{1/2}$ determines how quickly new information is incorporated into prices. For liquid Polymarket contracts:

$$\tau_{1/2}^{\text{Polymarket}} \approx \text{minutes to hours} \tag{10}$$

Our system operates at:

$$\tau_{\text{cycle}} \approx 6{,}917\text{s} \approx 1.9 \text{ hours (per cycle)} \quad \text{with 1 cycle/day} \tag{11}$$

The effective detection latency—from information emergence to gap flagging—is **12–24 hours**, far exceeding $\tau_{1/2}$. Any genuine gap has been closed by informed traders long before our system detects it.

7.2 No Information Advantage

The system reads the same public sources (Reddit, Bluesky, news RSS) that Polymarket participants read. It has no proprietary data, no insider information, and no access to non-public signals. Its only potential advantage is *processing speed*—running NLP on sources faster than human reading—but this advantage is negated by the daily cycle frequency.

7.3 Sentiment $\neq$ Information

Equation 4 assumes that aggregate sentiment maps linearly to probability via parameter α. This assumption fails for several reasons:

- Social media sentiment is **structurally biased positive** (fans, enthusiasts, and advocates post more than critics)
- Sentiment intensity does not correlate with information quality (a passionate fan post and an expert analysis receive equal weight)
- The mapping parameter $\alpha = 0.4$ is **not calibrated**—it was set heuristically, not derived from historical data

7.4 Insufficient Historical Data

With only 8 records in the `historical_odds` table, the pattern deviation detector (Type 3) and information asymmetry detector (Type 2) have inadequate data for statistical analysis. The z-score computation requires sufficient samples to estimate μ and σ reliably; with $n = 8$, confidence intervals are too wide for meaningful anomaly detection.

7.5 Liquidity Constraint

Even if a genuine gap were detected in an illiquid market, the cost of executing the trade (bid-ask spread + market impact) would likely exceed the edge. For the NHL longshot contracts where most gaps were detected, a single large order would move the price substantially, eliminating the edge before it could be captured.

8 What Would Be Required

Based on our analysis, a system capable of detecting genuine prediction market inefficiencies would require:

Table 7: Gap analysis: current system vs. requirements for viable gap detection.

Dimension	**Current System**	**Required System**
Cycle Frequency	1×/day (8 PM)	Every 5–15 minutes
Information Sources	Public social media	Event-driven feeds, filings, insider-adjacent data
Sentiment Calibration	Uncalibrated ($\alpha = 0.4$)	Source-specific calibration curves
Market Universe	625 contracts (broad)	10–20 liquid, catalyst-rich contracts
Validation	0 resolved predictions	30+ days of tracked resolutions
Backtesting	Framework exists, 0 data	3–6 months historical backtest
Historical Odds	8 records	Continuous tick-level price history
Execution	Analysis only	Integrated order placement

9 Alternative Application: Prediction Market Odds as Equity Signal Features

While our system fails to generate tradable signals *within* prediction markets, the infrastructure has a potentially valuable alternative application: using Polymarket odds as **input features for equity forecasting models**.

9.1 Theoretical Basis

Prediction market prices on policy events (Federal Reserve rate decisions, tariff announcements, geopolitical conflicts) encode the market's consensus probability of outcomes that directly affect equity valuations. These probabilities are:

- **Forward-looking:** Unlike economic indicators (lagging) or surveys (slow-updating), prediction market prices update in real-time as new information arrives.
- **Incentive-aligned:** Participants have financial stakes in accuracy, unlike survey respondents.
- **Continuously available:** Unlike quarterly earnings or monthly employment reports, prediction market odds are available 24/7.

9.2 Proposed Integration

Polymarket odds on macro-relevant events could serve as features in time-series forecasting models such as Temporal Fusion Transformers (TFT) (Lim et al., 2021). TFT models are particularly well-suited because they:

1. Handle mixed-frequency inputs (daily equity prices + real-time prediction market odds)
2. Provide interpretable attention weights showing *which* prediction market features drive equity forecasts
3. Support static (contract metadata), known future (event dates), and observed (odds history) input types natively

Example features: Fed rate cut probability, tariff escalation probability, ceasefire probability, election outcome odds—each representing a macro risk factor that traditional factor models cannot easily capture.

9.3 Why This May Succeed Where Direct Trading Failed

The key distinction is that equity markets are *not* directly pricing prediction market odds. While Polymarket participants rapidly incorporate information into prediction market prices, the *equity implications* of those probabilities may be underappreciated by stock market participants. This creates a potential information asymmetry that our data pipeline is well-positioned to exploit—not by trading prediction markets, but by using their odds as a novel feature class for equity models.

10 Conclusion

This paper presents a comprehensive multi-agent system for prediction market pricing gap detection, demonstrating both the engineering feasibility of large-scale sentiment-driven market analysis and the economic reality of attempting to outperform informationally efficient markets with public data at low frequency.

Our key findings are:

1. **The architecture works; the signal does not.** The system successfully collects, processes, and analyzes 32,791 social posts across 625 contracts using ensemble NLP, but the detected "gaps" reflect social media optimism bias rather than genuine market inefficiency.
2. **Polymarket is consistent with weak-form efficiency for liquid contracts.** The market incorporates public information faster than a daily-frequency analytical system can detect divergences.
3. **The speed gap is structural, not fixable by optimization.** Even reducing cycle time from 2 hours to 5 minutes would not close the gap with informed traders who monitor these markets continuously.

4. **Prediction market odds as equity features represent a more promising application.** The infrastructure built for this project—multi-source data collection, sentiment analysis, database persistence—can be repurposed to generate novel features for equity forecasting models.

The negative result itself is informative: it provides empirical evidence for prediction market efficiency from the perspective of an automated system attempting to exploit inefficiencies, complementing the theoretical literature with a practical engineering perspective.

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